Generator selection and CTGAN fills

Status of the two tasks: issue #56 (generator rules) and issue #55 (Default and Polish synthetic minority rows).

This note records what is finished, how each piece was built, the numbers from the files on disk, and the steps taken to produce them. Comparisons are the 50% fill against real minority rows in the cleaned training split. Validation and test were not read.

1. What is done

Both tasks have a working result. Generator selection is on main. Default and Polish synthetic files exist on the local branch *arrico/ctgan-polish*, in two versions.

Task 1. Generator selection (issue #56)

- The model-selection agent measures the cleaned training table and chooses a generator by the written rules.
- CTGAN is the only generator that trains. TimeGAN and WGAN-GP return SKIPPED, with the reason logged.
- The work is on main through pull request #67 (commit 592e2d3, merge 58d5896).

Task 2. Default and Polish CTGAN fills (issue #55)

- Each dataset has nested 10%, 25%, and 50% minority files. The smaller files are the first rows of the 50% file.
- Version 1 uses the original Bank training settings and conditional sampling. It is committed as 6f0c0fd.
- Version 2 uses Kyle’s Bank CTGAN v2 recipe from pull request #69. Those files are written and checked. They are not committed yet.
- Each run has a loss curve, per-column plots, a correlation figure, and a written note on the columns that struggled.

Dataset	Version	10%	25%	50%	Where
Default	v1	1,004	2,509	5,018	default_ctgan_fill*.csv
Default	v2	1,004	2,509	5,018	default_ctgan_v2_fill*.csv
Polish	v1	2,353	5,883	11,766	polish_ctgan_fill*.csv
Polish	v2	2,353	5,883	11,766	polish_ctgan_v2_fill*.csv

All six version-2 files, and the earlier version-1 files, live under data/interim/<dataset>/synthetic/. Row counts are round(fill × class gap). Polish at 50% is 11,766 because round(11,766.5) uses banker's rounding.

2. How it was done

Generator rules

The selector reads the training split only. It checks row count, missing values, and leakage, then tests whether the table is a real sequence: an entity key, a usable time index, at least five steps, and a repeated observation. If all four pass, the rule names TimeGAN for the monthly block and records CTGAN as the fallback. TimeGAN is not implemented, so the agent returns SKIPPED. If the temporal test fails and the table has categorical or indicator columns, the agent trains CTGAN. If every feature is numeric, the rule names WGAN-GP and the agent returns SKIPPED, with CTGAN as the documented fallback. A failed quality gate can add 100 epochs and try CTGAN once more. The decision log stores the rule, the inputs, and the outcome.

CTGAN files

Both runners call the shared CTGAN tool. They train on the cleaned training table after dropping source_row. Default’s target is y = 1 (3,982 of 18,000 rows; class gap 10,036). Polish’s target is bankrupt = 1 (1,255 of 26,043

rows; class gap 23,533). The Polish table is the pooled cohort file, used as provided. Discrete columns come from datasets.json: nine Default columns (SEX, EDUCATION, MARRIAGE, PAY_1 through PAY_6) and, for Polish, cohort only. Numeric columns are clipped to the training range and whole-number columns are rounded. Exact copies of real training rows are dropped.

Version 2 is the recipe in docs/notes/ctgan_bank_v2.md, stored as the v2 bundle in src/agentic_gan/utils/ctgan_settings.py. The run is `--settings v2`.

Setting	v1 (original Bank run)	v2 (pull request #69)
Epochs / batch / seed	300 / 500 / 42	300 / 500 / 42
Network widths	256, 256	256, 256
pac	10	5
Discriminator steps	1	3
Sampling	Conditional on the minority class	Unconditional; minority rows kept
Pool before the 50% file	The 50% file itself	60,000 minority rows
Raking	None	Discrete columns matched to real minority shares
Device	CPU	CPU

The Bank recipe also drops rows that break Bank contact-history rules (pdays, poutcome, previous). The shared rule list defines those rules for Bank Marketing only. Default and Polish have no documented rule that holds for every real training row, so no row filter was added.

Raking follows the Bank sampler: fifty passes that reweight the pool so each discrete column’s shares match the real minority rows, then a sample without replacement. On Bank the raked columns were job, month, poutcome, marital, and day-of-month bins. Here the raked columns are the dataset’s own discrete columns. Day bins were not invented for Default or Polish.

3. Current results

Format checks passed for both version-2 runs. Every synthetic row has the minority target and is_synthetic = 1. There is no source_row column, no exact duplicate row, and no exact copy of a real training row. Discrete codes all appear in the training data. Each loss file has 300 epochs. The 10% file is the first rows of the 25% file, and the 25% file is the first rows of the 50% file.

Default v2 trained in 1,066.6 seconds. The sampler drew 213,958 rows to keep 60,000 minority rows (29.0% of draws had y = 1). Polish v2 trained in 3,277.8 seconds (about 55 minutes) and then raked a 60,000-row minority pool down to 11,766 rows. Final losses: Default generator –0.346, discriminator –0.294; Polish generator 0.777, discriminator –0.357.

Default, fill 50% against real training rows with y = 1

Check	v1	v2
Training time	503.5 s	1,066.6 s
Mean KS, 14 numeric columns	0.150	0.128
Numeric columns with KS ≥ 0.15	7 of 14	5 of 14
Worst numeric KS	PAY_AMT3 0.232	PAY_AMT6 0.250
Mean TVD, 9 discrete columns	0.163	0.012

Check	v1	v2
Worst discrete TVD	SEX 0.355	PAY_6 0.021
Mean absolute correlation difference	0.078	0.053
PAY_1 delay share (real 52.3%)	51.7% synthetic	52.9% synthetic

Version 2 brings SEX, EDUCATION, and the payment-status codes onto the real default shares. PAY_1 through PAY_6 delay rates stay within about one percentage point of the real default rows. The payment-amount columns still struggle. PAY_AMT1 through PAY_AMT6 are too rarely zero (synthetic zero shares about 6–16%, against about 27–31% of real default rows). PAY_AMT6 is the worst numeric column at KS 0.250. Bill amounts are closer; AGE is close (KS 0.039).

Polish, fill 50% against real training rows with bankrupt = 1

Check	v1	v2
Training time	1,229.7 s	3,277.8 s
Mean KS, A1–A64	0.433	0.290
Columns with KS ≥ 0.15	56 of 64	52 of 64
Worst numeric KS	A27 0.952	A58 0.694
Cohort TVD	0.552	0.016
3year share (real 23.4%)	78.7% synthetic	22.2% synthetic
Mean absolute correlation difference	0.143	0.154

Version 2 cohort shares: 1year 11.5% (real 11.9%), 2year 19.9% (19.6%), 3year 22.2% (23.4%), 4year 25.3% (24.3%), 5year 21.2% (20.8%). The numeric ratios are still weak. The worst five are A58 (0.694), A37 (0.624), A54 (0.603), A45 (0.586), and A60 (0.580). The usual failure is a compressed spread. Several columns also flip sign: A64 is negative in 57% of synthetic rows and in 0% of real bankrupt rows; A37 is negative in 62.5% of synthetic rows and in 0.1% of real bankrupt rows.

Small discrete distances on version 2 are the raking step. Kyle’s Bank note makes the same point: matching the category shares is partly by design. The numeric KS and the correlation gap are the checks raking does not force. On Bank, version 2 still did not beat real-only training. That downstream comparison has not been run on these Default or Polish files.

4. What I did

- Read the generator-rules plan and wired select_generator into the model-selection agent, including SKIPPED for TimeGAN and WGAN-GP. That commit is 592e2d3 and is on main as pull request #67.
- Built eda/ctgan_default/ and eda/ctgan_polish/ runners and diagnose scripts from the Bank runner, pointed at each cleaned training table.
- Trained version 1 on both datasets (pac 10, one discriminator step, conditional sampling), checked the nested files, and wrote the plots and diagnostics. Committed that work as 6f0c0fd. Trained model files (.pkl) were left untracked, matching the Bank practice.
- Merged origin/main into arrico/ctgan-polish (d8550f5). The only conflict was the CTGAN sampling docstring. Main brought pull request #69, the Bank v2 note, the IEEE-CIS split, and the v1 quality note.
- Put the Bank v2 recipe into load_ctgan_settings("v2"): pac 5, three discriminator steps, unconditional sampling, and a 60,000-row pool that is raked. Tests cover that bundle.
- Retrained Default (17.8 minutes) and Polish (54.6 minutes) with --settings v2. Wrote default_ctgan_v2_fill{10,25,50}.csv and polish_ctgan_v2_fill{10,25,50}.csv beside the version-1 files, plus artifacts under eda/ctgan_*/artifacts_v2/.
- Re-ran both diagnose scripts on the version-2 fill-50 files. Version-1 diagnostics were left in the original artifacts folders.

Where the version-2 results sit

Item	Path
Default fills	data/interim/default/synthetic/default_ctgan_v2_fill50.csv
Polish fills	data/interim/polish/synthetic/polish_ctgan_v2_fill50.csv
Default note and plots	eda/ctgan_default/artifacts_v2/
Polish note and plots	eda/ctgan_polish/artifacts_v2/
Settings	src/agentic_gan/utils/ctgan_settings.py
Bank recipe	docs/notes/ctgan_bank_v2.md

The version-2 code, CSVs, plots, and diagnostics are local changes on arrico/ctgan-polish. They are not in a commit and not in a pull request. The comparison in issue #35 should use the version-2 fill-50 paths above. The unversioned default_ctgan_fill50.csv and polish_ctgan_fill50.csv files are still the version-1 pools.